

# Software Requirements Specification: Restaurant Order Processing and Inventory Management System

Group 24

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## Revision History

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# 1. Introduction

The purpose of this document is to define the software requirements for the **Restaurant Order Processing and Inventory Management System**. This system is intended to computerize the order processing, billing, accounting, and inventory management of a restaurant.

## 1.1 Purpose

This SRS document describes the goals, functionalities, and system requirements of the proposed restaurant management software. It outlines how the system will help restaurant owners streamline order processing, generate statistical reports, and maintain accurate inventory levels.

## 1.2 Scope

The system will allow restaurant staff to process customer orders efficiently, generate bills, and manage daily transactions. It will also track inventory levels and generate purchase orders when ingredient levels drop below a threshold. The software will be designed for a multi-user environment with access levels for managers, chefs, and sales clerks.

## 1.3 Definitions, Acronyms, and Abbreviations

- **POS** – Point of Sale
- **SKU** – Stock Keeping Unit
- **DBMS** – Database Management System

## 1.4 References

- Restaurant Management Guidelines
- Inventory Control Best Practices

## 1.5 Overview

The remaining sections of this document detail the overall system description, specific requirements, and supporting information.

## 2. Overall Description

### 2.1 Product Perspective

This system is a standalone restaurant management application designed to enhance efficiency in order processing, billing, and inventory tracking. It will interact with existing restaurant hardware, such as POS systems and barcode scanners, ensuring seamless integration into restaurant operations.

### 2.2 Product Functions

The system will have the following core functionalities:

- Order processing and billing
- Inventory management and stock tracking
- Automated purchase order generation
- Sales reporting and analytics
- Role-based access control
- Statistical sales reports for different menu items
- Automated invoice processing and cheque generation for supplier payments

### 2.3 User Characteristics

The system will be used by different user roles, including:

- **Managers:** Oversee operations, update pricing, and generate reports.
- **Sales Clerks:** Enter customer orders and process payments.
- **Chefs/Kitchen Staff:** View ingredient levels and request supplies.

### 2.4 Constraints

- The system must function on both desktop and mobile interfaces.
- It must support integration with third-party payment gateways.
- All financial transactions must be encrypted for security compliance.

## 2.5 Assumptions and Dependencies

- A stable internet connection is required for cloud-based features.
- Users must have basic knowledge of restaurant operations to use the system effectively.
- The restaurant will provide necessary hardware such as barcode scanners and receipt printers.

## 3. Specific Requirements

### 3.1 Functionality

- **Order Processing**
  - The system shall allow sales clerks to enter item codes and quantities sold.
  - The system shall generate itemized bills for customers.
  - The system shall maintain the prices of all items and allow managers to update them.
- **Inventory Management**
  - The system shall track ingredient stock levels.
  - The system shall generate purchase orders if stock levels fall below a computed threshold.
  - The system shall calculate the threshold based on the past three days' consumption and maintain at least two days' stock.
  - The system shall allow input of invoice data upon ingredient arrival, including quantity and price.
- **Billing and Accounting**
  - The system shall allow managers to update item prices.
  - The system shall generate monthly financial reports.
  - The system shall generate statistical reports for sales of different items.
  - The system shall automatically generate cheques for supplier payments when sufficient cash balance is available.
  - The system shall provide on-demand monthly sales and expense reports for managers.

### **3.2 Usability**

- The system shall provide an intuitive GUI for all users.
- The system shall allow role-based access for different staff members.

### **3.3 Performance**

- The system shall process an order within 2 seconds.
- The system shall support up to 100 concurrent users.

### **3.4 Security**

- The system shall encrypt all financial transactions.
- The system shall require login authentication for accessing sensitive data.

### **3.5 Supportability**

- The system shall support cloud-based deployment.
- The system shall allow integration with third-party payment gateways.

### **3.6 Design Constraints**

- The system shall be developed using a relational database.
- The system shall use standard web technologies.

### **3.7 Purchased Components**

- The system shall support the purchase and tracking of ingredients and inventory components.
- The system shall integrate with supplier databases to facilitate automated purchasing.
- The system shall store purchase history and invoice details for accounting purposes.
- The system shall generate purchase orders dynamically based on real-time inventory data.
- The system shall validate received stock against purchase orders to ensure accuracy.

### 3.8 Interfaces

- **User Interfaces:** The system shall provide an interactive GUI.
- **Software Interfaces:** The system shall integrate with existing accounting software.
- **Hardware Interfaces:** The system shall support barcode scanners for order processing.